Hybrid Learning Architecture Combining Deep Learning and Machine Learning for Multi-label Chest X-ray Classification

Prof.Priyanka Jadhav
Pimpri Chinchwad College of Engineering, Computer Science Engineering (AIML)
priyanka.jadhav@pccoepune.org

Vitthal Shahaji Humbe
Pimpri Chinchwad College of Engineering, Computer Science Engineering (AIML)
vitthal.3454@gmail.com

Vaibhavi Vivekanand Swami
Pimpri Chinchwad College of Engineering, Computer Science Engineering (AIML)
vibhaswami7@gmail.com

Vaishnavi Dagadu Dalave
Pimpri Chinchwad College of Engineering, Computer Science Engineering (AIML)
laxmiddalave@gmail.com

Samruddhi Manesh Dhumal
Pimpri Chinchwad College of Engineering, Computer Science Engineering (AIML)
samruddhidhumal028@gmail.com

***Abstract*— Interpreting chest X-rays remains a challenging task due to overlapping visual manifestations across thoracic diseases and the severe class imbalance inherent in large-scale medical datasets, which often biases models toward majority classes while underperforming on clinically critical but rare conditions. In this study, we address multi-label thoracic pathology classification using the NIH Chest X-ray dataset comprising 112,120 images annotated across 14 pathologies, and propose a hybrid architecture that explicitly decouples deep learning from downstream decision-making. First, a fine-tuned DenseNet-121, trained with targeted on-the-fly data augmentation, acts strictly as a deep feature extractor. The extracted 1024-dimensional representations are then classified using classical machine learning algorithms. Model performance is comprehensively evaluated using ROC-AUC, F1-score, recall, and Negative Predictive Value (NPV) to ensure equitable assessment across all classes and maintain diagnostic reliability. Results demonstrate that the DenseNet–Random Forest ensemble achieves the highest AUC of 0.8430, marginally outperforming the SVM (0.8357), indicating that the proposed hybrid framework effectively mitigates majority-class bias while providing a robust and clinically supportive multi-label screening solution.**

Keywords*— *Deep Learning, Transfer Learning, CNN, Chest X-ray, Hybrid Architecture, Random Forest, SVM, DenseNet121

I. INTRODUCTION

Lung diseases are a varied group of conditions that affect the respiratory system. They significantly contribute to global illness and death rates [1]. These conditions range from acute, life-threatening infections like pneumonia and tuberculosis (TB) to chronic issues such as Chronic Obstructive Pulmonary Disease (COPD) and different forms of lung cancer [1], [2]. The serious socioeconomic and public health challenges posed by these diseases make early and accurate diagnosis essential for effective patient management and for reducing negative health outcomes [3]. For infectious diseases like pneumonia, a quick and reliable diagnosis is crucial. Delays in starting treatment can lead to severe complications and higher death risk, especially for vulnerable populations [7]. Additionally, the complexities introduced by new respiratory threats, such as COVID-19, highlight the ongoing need for innovation in pulmonary diagnostics [2].

For many years, the standard method for screening and diagnosing thoracic diseases has been Chest X-ray (CXR) radiography [4]. In this traditional, human-centered approach, the diagnosis relies on the visual assessment of CXR images by trained radiologists or specialized clinicians [7]. They inspect the images for subtle signs of disease, such as specific opacities, fluid buildup, or structural changes. While this method remains the clinical standard, it has limitations due to its dependence on subjective human judgment. The diagnostic process can vary between different experts and may suffer from reduced accuracy because of clinician fatigue during busy shifts [7]. The complexity of the image data adds further challenges, often resulting in inconsistent results, especially for unclear or subtle disease signs. This situation reveals a major gap: the need for an objective, automated, and widely applicable diagnostic tool to support and standardize human interpretation. The Chest X-ray (CXR) is the most common, cost- effective, and readily available form of medical imaging worldwide [6]. As a type of projection radiography, it uses low doses of ionizing radiation to create a two-dimensional image of the internal structures of the chest, including the lungs, heart, and ribcage. Its simplicity and speed make it the primary tool for initial screenings. However, interpreting these images is complicated because three-dimensional anatomy is flattened into a single plane, leading to overlap and occlusion of structures. Often, a single CXR image shows signs of multiple, coexisting conditions (e.g., Pneumothorax along with Edema). This situation creates a multi-label classification challenge: the diagnostic system must evaluate 14 different disease labels at once, requiring a higher level of analytical precision than traditional methods provide [5].

To address the challenges of manual CXR interpretation— such as subjectivity, variability, and throughput limits— research has turned to Computer-Aided Diagnosis (CAD) systems that use Artificial Intelligence

(AI) [4]. Machine Learning (ML), a key part of AI, offers the framework to develop algorithms that learn complex patterns and make predictions from data without needing explicit programming [8]. Early work in ML employed traditional classification techniques like Support Vector Machines (SVM) and Random Forests. While these algorithms are highly effective at establishing decision boundaries, they possess a major limitation when applied directly to images: they depend on manual feature selection. An expert must painstakingly define and extract relevant visual features (e.g., texture measures, shape characteristics) from the CXR before the ML model can start the classification process. This manual step is time-consuming, requires deep knowledge, and limits the model's ability to detect subtle or intricate patterns, making it less effective for the variability found in raw medical image data [4].

The approach changed significantly with the introduction of Deep Learning (DL), a specialized form of ML that uses Convolutional Neural Networks (CNNs). CNNs differ from traditional ML by learning features automatically [1]. Their architecture consists of multiple convolutional and pooling layers, allowing the model to learn complex feature representations straight from the raw pixel input [7]. The effectiveness of DL is often enhanced through Transfer Learning, which fine-tunes robust, pre-trained models on large medical image datasets. Specifically, densely connected architectures like DenseNet have proven to be exceptionally powerful in this domain. By connecting each layer to every other layer in a feed-forward fashion, DenseNet maximizes feature reuse and effectively abstracts the high-dimensional, overlapping visual cues necessary to identify complex thoracic pathologies [4], [6].

However, utilizing pure end-to-end Deep Learning for multi-label classification introduces its own set of challenges, particularly when faced with Class Imbalance—a widespread statistical issue in the NIH Chest X-ray dataset where certain diseases are exceedingly rare [10]. When conventional CNNs are trained entirely end-to-end on imbalanced data, the final classification layers often overfit to the majority classes. This leads to a high rate of false negatives, low recall, and poor clinical reliability for critical minority diseases. While targeted on-the-fly data augmentation (applying geometric changes like rotation and shifting) is essential to increase the effective sample size and visual diversity of minority classes during the CNN's feature learning phase [5], deep neural network classifiers can still struggle to draw optimal, robust decision boundaries across 14 highly skewed classes.

To address both the feature extraction limitations of traditional ML and the classification vulnerabilities of pure Deep Learning on imbalanced datasets, this research proposes a Hybrid Architecture. We deliberately decouple the visual abstraction process from the final decision-making process. First, a fine-tuned DenseNet-121 CNN is utilized exclusively as a deep feature extractor, converting complex, raw X-ray pixels into robust, 1024-dimensional continuous feature vectors. In the second phase, these high-dimensional representations are fed into classical Machine Learning algorithms—specifically Random Forest and Support Vector Machine (SVM) classifiers—to execute the final multi-label prediction. By combining the spatial understanding of CNNs with the robust, non-linear classification strengths of classical ML ensembles, this hybrid approach aims to mitigate class imbalance, enhance overall diagnostic reliability, and maximize the Negative Predictive Value (NPV) required for safe clinical screening

II. LITERATURE REVIEW

In the research study done by M. Jasmine and Ketan Kotecha et al. [9], titled “Lung Diseases Detection Using Various Deep Learning Algorithms”, they use two datasets for Pneumonia and Tuberculosis diseases which contains frontal chest X-ray images and one dataset of lung CT-scan images for cancer detection. Various preprocessing and data augmentation techniques are used to prepare the data for their research. They proposed three different CNN (convolutional neural network) models, namely – sequential model which is CNN model with 5 convolutional layers stacked in a sequential order, functional model which allows complex connection between layers and is more flexible architecture, and pre-trained VGG-16 model used with transfer learning. For pneumonia detection, the proposed sequential model gave 98% accuracy with 98.55% of F1-score and 96.33% of recall. For tuberculosis detection, the sequential model has 99% accuracy, 98% F1-score and 98.88% of recall. The functional model performed well for the cancer detection task in contrast to other models with 99.9% accuracy. They concluded that their proposed models are better in performance than other existing models.

In the study of Adnane Ait Nasser and Moulay A. Akhloufi [10], the author proposed the deep research on the publicly available chest X-ray radiography (CXR) datasets – Multi-disease datasets, Tuberculosis (TB) dataset, Pneumonia datasets, and Covid-19 datasets. They suggest that the key challenge lies in the data preprocessing and data preparation – how we are going to prepare the data for training and testing the Deep learning models. Advanced image processing techniques such as Generative Adversarial Networks (GANs), rotations, flipping, and scaling methods for data augmentation process, histogram equalization (HE), contrast-limited adaptive histogram equalization (CLAHE) and unsharp masking for image enhancement , and it also involves isolating the region of interest (ROI) for segmentation which helps model to focus on relevant areas of image, and other image processing techniques are used and reviewed by them. The model comprehensively studies various deep learning models for the detection of specific chest diseases like – custom CNNs, swin transformers, ResNet50 shows higher accuracy for pneumonia detection. DL models such as ResNet, RetinaNet show improved lung cancer detection performance. Custom DCNNs, VGG-16 and ensembles of models like AlexNet and GoogleNet have achieved higher accuracy on TB detection. For Covid-19 pandemic, Inception- V4 mobileNet-V2 and VGG-19 were adopted which shows highest accuracy of 95% and for multilabel classification on dataset which contains multiple labels per images models such as DenseNet-121 and Xception have been used to detect 14 different lung-related diseases simultaneously. This review highlights several challenges related to the datasets they reviewed such as – many large datasets are based on automated labelling from text reports which may introduce errors, datasets are also imbalanced which may lead to high accuracy but with high bias towards majority class, models often perform worse when they get tested on external data from different institutes, and most of the Deep Learning models acts as a “black box” which makes it difficult to understand their decision making process

In the research study by Aijaz Ahmad Reshi [5], the

authors proposed a deep CNN model to detect COVID-19 from chest X-ray images. The main challenge was the small and imbalanced dataset of X-ray images. To overcome this, they applied preprocessing steps such as dataset balancing, medical experts' image verification, and data augmentation (flipping and rotations) to improve data quality and size. The final CNN model contained 38 layers, including 6 convolutional layers, pooling, dropout, and fully connected layers, trained on 800 X-ray images (400 COVID-19 and 400 normal). It achieved 100% accuracy on test data and 99.5% accuracy on an independent dataset, showing very high reliability. Comparative analysis with other machine learning models (Random Forest, SVM, Logistic Regression, KNN, etc.) showed that the proposed CNN model performed better, especially on independent testing. The study concludes that CNN with extra convolutional layers and sufficient augmented data can be highly effective for automatic, fast, and low-cost COVID-19 detection using X-ray images, making it suitable for large-scale diagnosis during pandemics.

In the research study by Marriam Nawaz [6], titled "CXray-EffDet: Chest Disease Detection and Classification from X-ray Images Using the EfficientDet Model", the authors proposed a lightweight and efficient deep learning model for detecting and classifying chest diseases from X-ray images. The model, named CXray-EffDet, is based on the EfficientDet-D0 framework with EfficientNet-B0 as the backbone and is designed to detect diseased regions using bounding box annotations. The study used the NIH Chest X-ray dataset, which contains over 112,000 images across 14 disease categories, focusing on eight common chest diseases such as pneumonia, atelectasis, cardiomegaly, effusion, infiltration, mass, nodule, and pneumothorax. Preprocessing included bounding box labeling and performance evaluation with metrics such as Intersection over Union (IoU), mean Average Precision (mAP), accuracy, precision, recall, and F1- score. The proposed model achieved an IoU of 0.834, AUC of 0.908, average accuracy of 94.53%, precision of 90%, recall of 92.36%, and F1-score of 91.16%, outperforming existing CNN architectures like AlexNet, VGG16, ResNet50, and GoogLeNet. The study concludes that CXray-EffDet offers a robust, computationally efficient, and accurate approach for automated chest disease detection and classification, making it highly suitable for real-world clinical use.

In the paper "Pneumonia Detection Using CNN based Feature Extraction", et al. [7], the authors proposed a system to automatically diagnose pneumonia from chest X-rays by combining deep learning with machine learning classifiers. Their methodology involves using pre-trained Convolutional Neural Networks (CNNs), such as DenseNet and ResNet, as feature extractors and then feeding these features into a separate classifier for the final diagnosis. Through a series of experiments comparing different CNNs and classifiers on the ChestX-ray14 dataset, they found that the best result was achieved by using the DenseNet-169 model to extract features, which were then classified by a fine-tuned Support Vector Machine (SVM). This final customized model achieved an Area Under the Curve (AUC) score of 0.8002, demonstrating an effective approach for the pneumonia detection task.

In the research paper "Deep Convolutional Neural Networks for Chest Diseases Detection", et al. [1], the authors investigate the use of deep learning to classify pathologies in chest X-rays. Their methodology centers on designing a Convolutional Neural Network (CNN) to diagnose 12 common thoracic diseases, and for comparison, they also implement Traditional Back-Propagation Neural Networks (BPNNs) and Competitive Neural Networks (CpNNs). The networks were trained and tested using a large dataset from the National Institutes of Health Clinical Center. The results demonstrated the superiority of the deep learning approach; the proposed CNN achieved the highest recognition rate with 92.4% accuracy on the validation dataset, outperforming the best BPNN (89.57%) and CpNN (84.71%). The authors conclude that despite requiring a longer training time, the CNN's deep structure enables more effective feature extraction, leading to significantly better generalization and higher accuracy compared to the other models.

The research conducted by Rahman, T., Chowdhury [3], focuses on detecting pneumonia using chest X-ray images with the help of deep learning. Pneumonia, caused by bacteria or viruses, can be life-threatening if not diagnosed early. The study used four pre-trained CNN models (AlexNet, ResNet18, DenseNet201, and SqueezeNet) with a dataset of over 5,000 chest X-rays. The models were trained to classify normal, bacterial, and viral pneumonia cases. Among them, DenseNet201 achieved the highest accuracy, reaching up to 98% for normal vs pneumonia detection. This work shows that computer-aided systems can help doctors diagnose pneumonia faster and more accurately, especially in areas with fewer medical experts. This study talks about using artificial intelligence (AI), especially deep learning with CNNs, to detect pneumonia from chest X-ray images. Pneumonia is a major health problem worldwide, causing millions of deaths every year, especially in children. The authors used transfer learning, which means using pre-trained models that already learned from large datasets and applying them to medical images. Four popular models (AlexNet, ResNet18, DenseNet201, and SqueezeNet) were tested on more than 5,000 X-ray images. The images included normal lungs, bacterial pneumonia, and viral pneumonia. To improve results, the researchers used image preprocessing and data augmentation (like rotating and resizing images). Their results showed very high accuracy: 98% for detecting pneumonia, 95% for separating bacteria from viral, and 93.3% for identifying all three classes together. Among all models, DenseNet201 performed the best. This research proves that deep learning can help doctors by providing quick, reliable support for diagnosis, reducing errors, and saving time, especially in countries where expert radiologists are limited.

In the paper "Artificial Intelligence‑Based Classification of Chest X‑Ray Images into COVID-19 and Other Infectious Diseases", et al. [2], the authors developed an AI-based system for the rapid screening of infectious diseases from chest X-rays. Their approach utilized transfer learning combined with extensive data augmentation, creating 25 different types of augmented images from an original curated dataset to train generalized models. The methodology involved training and testing multiple classification models on datasets of adult, posteroanterior (PA) view X-rays corresponding to COVID-19, non-COVID-19, pneumonia, tuberculosis, and normal cases. The study's main result was that a combination of their two best-performing models, one trained on images rotated by 120° and another on images rotated by 140°, yielded the highest prediction accuracy across the different disease classes.

Authors Theodora Sanida, Maria Vasiliki Sanida, Argyrios Sideris, and Minas Dasygenis developed an

optimized deep learning model to improve the categorization of various lung conditions from chest X-ray images et al. [4]. Their approach was to enhance the well-known VGG16 architecture by integrating Squeeze-and-Excitation (SE) blocks, a technique that allows the model to dynamically focus on the most important features within an image. For their methodology, the researchers used a large dataset of chest X- rays covering six categories: lung opacity, normal, COVID- 19, bacterial pneumonia, viral pneumonia, and tuberculosis. They applied data augmentation techniques like rotation and scaling to balance the dataset and prevent model overfitting. The result was that their proposed VGG16-SE model showed superior performance compared to the classic VGG16, achieving an accuracy of 98.49%, an F1-score of 98.23%, a precision of 98.41%, and an accuracy of 98.86%.

The study conducted by Ibrahim D. M. [11], the authors introduces Deep-Chest, a deep learning framework designed to classify chest diseases—COVID-19, pneumonia, and lung cancer—along with normal cases, using both chest X-ray and CT images. Since X-rays may miss early signs while CT scans capture details before symptoms, combining both improves accuracy. The authors experimented with four models: VGG19+CNN, ResNet152V2, ResNet152V2+GRU, and ResNet152V2+Bi-GRU. After training on over 75,000 images, results showed VGG19+CNN outperformed others, achieving 98.05% accuracy, 98.43% precision, and 99.66% AUC. The work highlights deep learning's role in early, reliable detection of chest diseases, offering a single, unified model to reduce diagnosis time and errors. Future improvements may include larger datasets and advanced augmentation methods.

III. PROPOSED METHODOLOGY

A. Dataset : This study utilizes the publicly available NIH ChestX-ray14 dataset, comprising 112,120 frontal-view X-ray images from 30,805 unique patients. The dataset is fundamentally multi-label, meaning a single image can present multiple co-occurring thoracic conditions. It includes 14 distinct disease labels (e.g., Atelectasis, Cardiomegaly, Effusion) and a 15th "No Finding" class for scans with no detectable abnormalities.

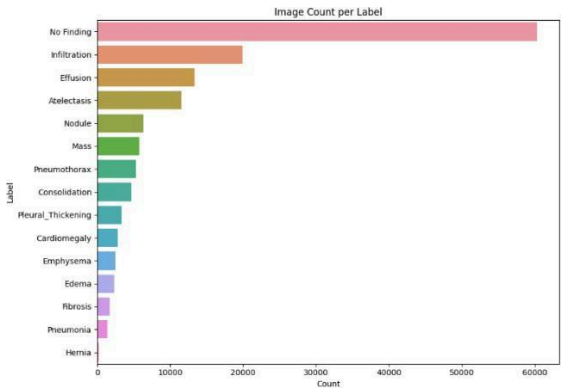

Fig 1. Class distribution of the 14 thoracic pathologies and the "No Finding" class within the NIH ChestX-ray14 dataset, illustrating severe class imbalance.

The primary challenge of this dataset is its severe, long-tailed class imbalance, which mirrors real-world clinical frequencies. As illustrated in the class distribution plot (Figure X), common pathologies like Infiltration are heavily overrepresented with tens of thousands of instances, while rare conditions like Hernia possess fewer than 300 samples. This extreme skew inherently biases conventional classifiers toward majority classes, necessitating the specialized data augmentation and hybrid extraction architecture proposed in this study.

B. Deep Learning for Feature Extraction

The initial phase of the proposed architecture focuses on translating high-dimensional, raw radiographic pixels into a lower-dimensional, highly discriminative feature space. Rather than relying on manual feature engineering, this phase leverages a Convolutional Neural Network (CNN) to automatically learn the spatial hierarchies of thoracic pathologies.

1) Data Preprocessing : Before feature extraction, standardizing the input space is mathematically necessary to ensure network convergence and numerical stability. All raw Chest X-ray images, originally of varying dimensions, are bilinearly interpolated to a fixed spatial resolution of 224×224 pixels. Following spatial resizing, pixel intensity normalization is applied across the RGB channels. Each pixel xi,j is normalized using the mean μ and standard deviation σ derived from the ImageNet dataset:

$$x'_{i,j} = \frac{x_{i,j} - \mu}{\sigma}$$

This zero-mean, unit-variance standardization ensures that the pre-trained weights of the CNN do not encounter severe distribution shifts when processing the medical images.

2) Targeted Data Augmentation Strategy: To combat the severe class imbalance inherent in the NIH dataset, a bifurcated data augmentation pipeline is implemented. During the training of the CNN, majority class images bypass augmentation via a non-augmentation interface to prevent the over-representation of common features. Conversely, minority class images undergo on-the-fly stochastic data augmentation (e.g., rotations, horizontal flips, and cutouts) to artificially expand their spatial diversity. Geometrically, spatial transformations such as rotation by an angle θ are achieved using an affine transformation matrix mapped to the original pixel coordinates (x,y):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

By applying these transformations exclusively to underrepresented classes during the gradient descent optimization, the network is forced to learn robust, invariant features for rare diseases rather than memorizing their limited spatial orientations.

3) Transfer Learning via DenseNet-121 For the core feature extraction engine, this study utilizes the DenseNet-121 architecture, initialized with weights pre-trained on the ImageNet database. DenseNet was specifically selected for its ability to mitigate the vanishing gradient problem and maximize feature reuse—critical traits when identifying the subtle, overlapping textures of multiple thoracic diseases. Unlike traditional CNNs where the l-th layer only receives the feature maps from the (l−1)-th layer (i.e., xl=Hl(xl−1)), DenseNet introduces a dense connectivity pattern. In a dense block, the l-th layer receives the concatenated feature maps of all preceding layers:

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}])$$

where […] denotes the concatenation operation along the channel dimension, and the non-linear transformation Hl(·) is a composite function of Batch Normalization (BN), Rectified Linear Unit (ReLU) activation, and Convolution (3×3). This dense connectivity allows the network to preserve low-level visual features (such as sharp rib edges or localized opacities) all the way to the deepest layers of the network.

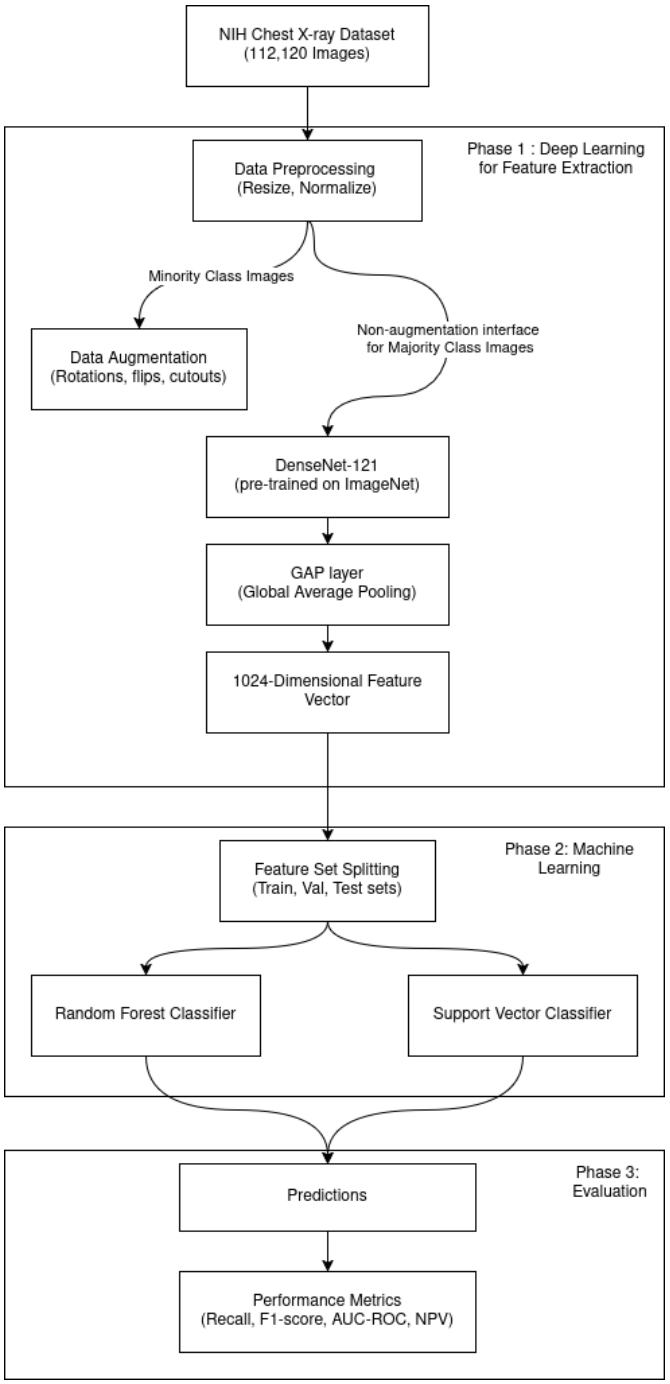

fig. 2: Proposed hybrid methodology architecture detailing the data augmentation pipeline, DenseNet-121 feature extraction, and multi-label machine learning classification.

4) Global Average Pooling (GAP) : Layer In standard classification tasks, the final convolutional feature maps are flattened and passed through a fully connected dense layer (the classification head). In our hybrid methodology, this classification head is deliberately detached. Instead, the final outputs are passed through a Global Average Pooling (GAP) layer. Assuming the final convolutional block of DenseNet-121 outputs a feature map tensor of dimensions H×W×C (where Height H=7, Width W=7, and Channels C=1024), the GAP layer computes the spatial average of each feature channel c. The resulting scalar value Fc for a single channel is calculated as:

$$F_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} f_c(i,j)$$

where fc(i,j) represents the activation at spatial coordinate (i,j) of the c-th channel. This operation drastically reduces the parameter count and minimizes spatial overfitting.

5) Final Outcome: 1024-Dimensional Feature Vector. The direct output of the GAP layer is a continuous, 1-dimensional array. For the DenseNet-121 backbone, this results in a fixed-size vector V∈R1024. This 1024-dimensional feature vector represents a highly compressed, mathematically dense abstraction of the patient's entire chest radiograph. By converting 40 gigabytes of complex image pixels into these lightweight tabular vectors, these phase successfully constructs the optimal, low-noise input representation required to train the classical Machine Learning classifiers

C. Machine Learning Classification

Following the deep feature extraction phase, the diagnostic problem transitions from spatial image analysis to high-dimensional tabular classification. The input to Phase 2 is the globally pooled feature matrix $X \in R^{N \times d,}$ where N is the number of X-ray images and d=1024 represents the extracted DenseNet-121 feature dimensions. Because the NIH dataset requires multi-label classification (predicting the presence or absence of 14 independent pathologies simultaneously), the target space is defined as $Y \in \{0,1\}^{N \times 14}$. To solve this, the 1024-dimensional vectors are evaluated using two distinct classical machine learning algorithms: a Support Vector Machine (SVM) and a Random Forest (RF) ensemble.

1) Multi-Output Support Vector Machine (SVM) : The first classification approach utilizes a linear Support Vector Machine. Given the high-dimensional, highly abstracted nature of the 1024-dimensional CNN features, a complex non-linear kernel (such as RBF) is computationally prohibitive and empirically unnecessary. Instead, a Linear SVC is employed to find the optimal maximum-margin hyperplanes. Because standard SVMs are inherently binary classifiers, the multi-label problem is mathematically formulated using a One-vs-Rest (Multi-Output) strategy. For each pathology k∈{1,2,…,14}, the SVM solves the following primal soft-margin optimization problem:

$$\min_{w_k, b_k, \xi} \frac{1}{2}||w_k||^2 + C \sum_{i=1}^{N} \xi_i^{(k)}$$

Subject to the constraints:

$$y_i^{(k)}(w_k^T x_i + b_k) \geq 1 - \xi_i^{(k)}, \quad \xi_i^{(k)} \geq 0$$

where wk is the weight vector for the k-th pathology, bk is the bias, C is the regularization parameter controlling the penalty for misclassification, and ξi(k) are the slack variables allowing for non-separable data points. To evaluate clinical

metrics such as ROC-AUC, the raw decision boundary distances generated by the linear SVM must be converted into valid probability scores P(y(k)=1|x). This is achieved through Platt Scaling, which fits a logistic regression model to the SVM scores fk(x):

$$P(y^{(k)} = 1|x) = \frac{1}{1 + \exp(Af_k(x) + B)}$$

where parameters A and B are optimized using maximum likelihood estimation on the calibration data.

2) Random Forest Ensemble Classifier: The second classification architecture evaluated is a Random Forest, an ensemble learning method constructed from a multitude of decision trees. Random Forests are inherently robust to feature scaling and excel at capturing complex, non-linear interactions between the 1024 CNN features without requiring explicit kernel transformations.

During training, T independent decision trees are grown using bootstrap aggregating (bagging). For each node split within a tree, a random subset of features is evaluated to minimize the Gini Impurity, defined for a given node as:

$$I_G(p) = 1 - \sum_{c=0}^{1} p_c^2$$

where p_c is the fraction of samples belonging to class c within that node.

For the multi-label inference of a new X-ray feature vector x, the Random Forest aggregate probability for a specific pathology k is calculated by averaging the probabilistic predictions of all individual trees:

$$\hat{P}(y^{(k)} = 1|x) = \frac{1}{T} \sum_{t=1}^{T} P_t(y^{(k)} = 1|x)$$

Where $\hat{P}(y^{(k)} = 1|x)$ represents the estimated probability output by the t-th decision tree. This ensemble averaging drastically reduces model variance and helps prevent overfitting to the majority classes present in the imbalanced training set.

D. Model Evaluation and Clinical Metrics

Evaluating a multi-label classification model on an imbalanced medical dataset requires specialized metrics. Relying strictly on global accuracy is deeply misleading, as a naive classifier could achieve high performance simply by predicting the majority class (e.g., "No Finding") while completely missing critical, rare pathologies. To rigorously validate the diagnostic reliability of the hybrid architectures, performance is evaluated using Macro-averaged class metrics, F1-Score, and Negative Predictive Value (NPV). For a given pathology $k \in \{1,2,\ldots,14\}$, the foundation of all metrics relies on the elements of the confusion matrix: True Positives (TP_k), True Negatives (TN_k), False Positives (FP_k), and False Negatives (FN_k).

1) Macro-Averaging Strategy : To ensure that performance on rare minority classes (such as Hernia) is not mathematically overshadowed by massive majority classes (such as Infiltration), this study utilizes Macro-averaging. A macro-averaged metric calculates the unweighted arithmetic mean of the metric across all K=14 pathologies:

$$\text{Macro-Metric} = \frac{1}{K} \sum_{k=1}^{K} \text{Metric}_k$$

This structural choice forces the model to be evaluated equally on its ability to detect both common and exceedingly rare diseases.

2) Precision, Recall (Sensitivity), and F1-Score : In the context of computer-aided diagnosis, missing a diseased patient is far more dangerous than triggering a false alarm. Therefore, measuring Recall (Sensitivity)—the proportion of actual positive cases correctly identified—is crucial:

$$\text{Recall}_k = \frac{TP_k}{TP_k + FN_k}$$

Conversely, Precision measures the confidence in a positive prediction (the proportion of predicted positive cases that are truly positive):

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}$$

Because there is an inherent mathematical trade-off between Precision and Recall, the models are evaluated using the F1-Score. The F1-Score is the harmonic mean of Precision and Recall, providing a single, balanced metric that heavily penalizes models that blindly predict one class to exploit the data imbalance:

$$F1_k = 2 \times \frac{\text{Precision}_k \times \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}$$

3) Negative Predictive Value (NPV): While F1-score measures disease detection, the primary goal of many automated radiographic systems is clinical triage—safely filtering out healthy patients to save radiologist bandwidth. This is mathematically captured by the Negative Predictive Value (NPV), which defines the probability that a patient genuinely does not have the disease when the model predicts a negative outcome:

$$NPV_k = \frac{TN_k}{TN_k + FN_k}$$

A high NPV is the gold standard for clinical safety, as it proves the AI minimizes dangerous False Negatives and can be safely utilized as a frontline screening tool.

4) Area Under the Receiver Operating Characteristic Curve (ROC-AUC): Metrics like F1 and NPV depend on a strict binarization threshold (typically 0.5). To evaluate the fundamental discriminative capability of the hybrid models independent of this threshold, the ROC-AUC is calculated. The ROC curve plots the True Positive Rate (Recall) against the False Positive Rate across all possible probability cutoffs. The AUC represents the definite integral under this curve:

$$AUC_k = \int_0^1 \mathrm{TPR}_k(\mathrm{FPR}_k)\, d(\mathrm{FPR}_k)$$

An AUC of 1.0 indicates perfect separation between diseased and healthy X-rays, while an AUC of 0.5 represents a random guess. The Macro-AUC serves as the ultimate benchmark for comparing the deep spatial understanding translated by the DenseNet-121 feature vectors into the Random Forest and SVM classifiers.

IV. RESULTS & CONCLUSION

The primary objective of this study was to evaluate the efficacy of decoupling deep spatial feature extraction from multi-label decision-making. Performance was measured across 14 distinct thoracic pathologies using Macro-averaged ROC-AUC to ensure minority classes were not eclipsed by majority distributions. The hybrid DenseNet-121 and Random Forest ensemble achieved a superior overall class separability, recording a Macro ROC-AUC of 0.8430. The DenseNet-121 and Support Vector Machine (SVM) configuration followed closely with a Macro ROC-AUC of 0.8357. The Random Forest's slight superiority indicates that the bagging ensemble approach was highly effective at modeling the complex, non-linear interactions within the 1024-dimensional feature space generated by the CNN's Global Average Pooling layer.

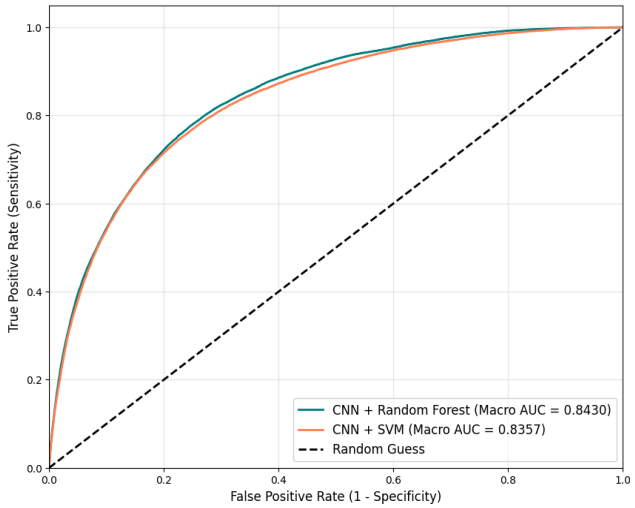

fig. 3 : Macro-average ROC curves comparing the DenseNet-121 + Random Forest ensemble versus the DenseNet-121 + SVM configuration.

A. Class Imbalance and Threshold Metrics : While ROC-AUC evaluates the models across all possible probability thresholds, clinical deployment requires a fixed threshold (e.g., 0.5) to generate binary diagnostic flags. At this strict threshold, an interesting divergence in model behavior was observed.

Despite having a lower overall AUC, the SVM demonstrated significantly better resilience to class imbalance regarding positive disease detection. The SVM achieved a Macro F1-Score of 0.1982 and a Macro Recall of 0.1372, outperforming the Random Forest (F1: 0.1415, Recall: 0.0889). The Random Forest acted overly conservative, prioritizing precision and minimizing false alarms at the steep cost of missing positive cases. Conversely, the SVM's maximum-margin hyperplanes proved much more capable of correctly flagging minority-class instances, exhibiting F1-scores for rare conditions like Atelectasis (0.2183) and Pneumothorax (0.3195) that were double those of the Random Forest.

B. Clinical Viability and Negative Predictive Value (NPV)

The most significant clinical finding of this hybrid architecture is the remarkably high Negative Predictive Value (NPV) maintained across the disease spectrum. Both models consistently achieved NPVs exceeding 0.90 for the vast majority of pathologies. For critical conditions such as Hernia and Cardiomegaly, the SVM recorded NPVs of 0.9971 and 0.9640, respectively.

In a real-world triage setting, a high NPV is the ultimate benchmark for safety. These results suggest that if the proposed AI predicts a patient does *not* have Cardiomegaly, the attending physician can be over 96% confident in that negative assessment. While false positives remain a challenge due to the extreme dataset imbalance, the hybrid architecture has proven to be a highly reliable primary screening tool capable of safely ruling out severe conditions to optimize radiologist workflows.

C. Conclusion

This study addressed the profound challenges of multi-label classification and severe class imbalance in the automated diagnosis of thoracic diseases from Chest X-rays. By proposing a novel hybrid architecture, we successfully decoupled spatial visual abstraction from the final predictive algorithm. Utilizing a fine-tuned DenseNet-121 strictly as a feature extractor—combined with targeted on-the-fly data augmentation—the system effectively translated complex radiographic images into robust 1024-dimensional representations.

When applied to classical machine learning classifiers, this methodology proved highly effective. The DenseNet-Random Forest pipeline achieved a leading Macro ROC-AUC of 0.8430, while the DenseNet-SVM pipeline demonstrated superior F1-scores for minority classes. Most importantly, the architecture maintained exceptional Negative Predictive Values (>0.90) across critical pathologies. This proves that integrating deep feature extraction with traditional machine learning ensembles is a computationally efficient, highly reliable approach to mitigating majority-class bias and developing safe, frontline medical screening tools.

	Random Forest			***Support Vector Machine***		
Pathology	***AUC***	***F1-Score***	***NPV***	***AUC***	***F1-Score***	***NPV***

Atelectasis	0.7472	0.1091	0.8779	0.7476	0.2183	0.8857
Cardiomegaly	0.8413	0.1776	0.9623	0.8371	0.2280	0.9640
Consolidation	0.7141	0.0000	0.9291	0.6891	0.0462	0.9302
Edema	0.8172	0.0697	0.9651	0.8007	0.1352	0.9666
Effusion	0.8128	0.3642	0.8539	0.8117	0.4231	0.8643
Emphysema	0.8887	0.2493	0.9636	0.8646	0.3075	0.9662
Fibrosis	0.7597	0.0349	0.9833	0.7677	0.0596	0.9835
Hernia	0.7970	0.2617	0.9972	0.8803	0.2476	0.9971
Infiltration	0.6919	0.2010	0.7767	0.6842	0.3081	0.7909
Mass	0.7947	0.2084	0.9396	0.7894	0.2540	0.9423
Nodule	0.7353	0.067	0.9386	0.7354	0.1455	0.9414
Pleural Thickening	0.7358	0.0069	0.9555	0.7376	0.0728	0.9569
Pneumonia	0.6638	0.0035	0.9783	0.5856	0.0099	0.9784
Pneumothorax	0.8495	0.2273	0.9083	0.8273	0.3195	0.9159

Table 1: Per-Class Performance Metrics (F1-Score, NPV, and AUC) across 14 Thoracic Pathologies.

D.Future Work: Future iterations of this research will focus on implementing dynamic, class-specific probability thresholds to further optimize the trade-off between Precision and Recall for extreme minority classes. Additionally, the extracted 1024-dimensional feature vectors could be utilized to cluster similar patient profiles, paving the way for Content-Based Image Retrieval (CBIR) systems that provide radiologists with historical cases visually similar to the scan they are currently diagnosing.

V. REFERENCES

[1] Abiyev, Rahib H., and Mohammad Khaleel Sallam Ma'aitah. "Deep Convolutional Neural Networks for Chest Diseases Detection." Journal of Healthcare Engineering, vol. 2018, 1 Aug. 2018, pp. 1–11, www.hindawi.com/journals/jhe/2018/4168538/, https://doi.org/10.1155/2018/4168538.

[2] Sharma, A., Rani, S., & Gupta, D. (2020). Artificial Intelligence-Based Classification of Chest X-Ray Images into COVID-19 and Other Infectious Diseases. International Journal of Biomedical Imaging, 2020, 1–10. https://doi.org/10.1155/2020/8889023

[3] Rahman, T., Chowdhury, M. E. H., Khandakar, A., Islam, K. R., Islam, K. F., Mahbub, Z. B., Kadir, M. A., & Kashem, S. (2020). Transfer Learning with Deep Convolutional Neural Network (CNN) for Pneumonia Detection Using Chest X-ray. Applied Sciences, 10(9), 3233. https://doi.org/10.3390/app10093233

[4] Sanida, T.; Sanida, M.V.; Sideris, A.; Dasygenis, M. Optimizing Lung Condition Categorization through a Deep Learning Approach to Chest X-ray Image Analysis. BioMedInformatics 2024, 4, 2002–2021.https://doi.org/10.3390/biomedinformatics4030109

[5] Reshi, A. A., Rustam, F., Mehmood, A., Alhossan, A., Alrabiah, Z., Ahmad, A., Alsuwailem, H., & Choi, G. S. (n.d.). An Efficient CNN Model for COVID‐19 Disease Detection Based on X‐Ray Image Classification. Complexity, 2021(1). https://doi.org/10.1155/2021/6621607

[6] Nawaz, M.; Nazir, T.; Baili, J.; Khan, M.A.; Kim, Y.J.; Cha, J.-H. CXray-EffDet: Chest Disease Detection and Classification from X-ray Images Using the EfficientDet Model. Diagnostics 2023, 13, 248. https://doi.org/10.3390/diagnostics13020248

[7] Varshni, D.; Nijhawan, R.; Thakral, K.; Mittal, A.; Agarwal, L. Pneumonia Detection Using CNN Based Feature Extraction. *IEEE*, 2019. Available online: https://ieeexplore.ieee.org/document/

[8] N. Kühl and M. Goutier, "Machine Learning in Artificial Intelligence: Towards a Common Understanding," in Proc. 52nd Hawaii Int. Conf. Syst. Sci., 2019

[9] Priyadarsini, M. J. P., Kotecha, K., Rajini, G. K., Hariharan, K., Raj, K. U., Ram, K. B., Indragandhi, V., Subramaniyaswamy, V., & Pandya, S. (2023). Lung diseases detection using various deep learning algorithms. Journal of Healthcare Engineering, 2023, 1–13. https://doi.org/10.1155/2023/3563696

[10] Ait Nasser, A.; Akhloufi,M.A. A Review of Recent Advances in Deep Learning Models for Chest Disease Detection Using Radiography. Diagnostics 2023, 13, 159. https://doi.org/10.3390/diagnostics13010159

[11] Ibrahim, D. M., Elshennawy, N. M., & Sarhan, A. M. (2021). Deep- chest: Multi-classification deep learning model for diagnosing COVID-19, pneumonia, and lung cancer chest diseases. Computers in Biology and Medicine, 132, 104348. https://doi.org/10.1016/j.compbiomed.2021.104348